

Solution Requirements

Project Name: CaptionAI: AI-powered social media caption generator

Date: 7 July 2026

Team Name: Anupriya

Maximum Marks: 4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system must allow users to securely register, log in, and reset passwords via standard email/password credentials or OAuth 2.0 single sign-on (Google/LinkedIn).	High
2	Authorization levels	The platform will support two distinct tiers: Free Tier Users (access to standard generation and 10 credits/month) and Premium Tier Users (unlimited multi-tone text generation and history export access).	High
3	External interfaces	The system must connect to third-party Large Language Model APIs (e.g., OpenAI API) to exchange user context payloads and receive raw generated text data strings securely.	High
4	Transactions processing	The platform must securely process payment transactions via a centralized gateway integration (e.g., Stripe) and safely deduct generation credits in real-time.	Medium

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
5	Reporting	The system must generate a basic metrics dashboard displaying credit consumption, historical copy trends, and export character limits for user review.	Medium
6	Business rules, etc.	The core generator workflow must systematically strictly enforce specific platform character counts (e.g., maximum 2,200 characters for Instagram captions).	High
7	Compliance to laws	The architecture must adhere to global data privacy laws (e.g., GDPR) to safeguard user email logs and account payment details.	High
8	Other	Provide a localized "Copy to Clipboard" single-click function for text containers to enable rapid deployment to external mobile applications.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Ensure the system processes text formatting payloads and displays standard results promptly.	API response time must remain under 3 seconds per raw prompt payload generation loop.
2	Scalability	The backend environment must effortlessly autoscale to accommodate sudden influxes of concurrent active traffic without drops in performance.	Comfortably handle up to 5,000 concurrent active generation sessions simultaneously.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
3	Security & Data Privacy	Protect user credentials and system history logs from unauthorized external access queries.	Complete encryption of all traffic in transit via SSL/TLS and storage of user passwords using bcrypt hashing.
4	Reliability & Availability	The text generation platform must remain functional and accessible around the clock for business use.	Guarantee a minimum application availability of 99.5% uptime calculated monthly.
5	Usability & Accessibility	The frontend UI interface must be intuitive, clean, and entirely operational across multiple viewport breakpoints.	Achieve full responsiveness on mobile web layouts and pass basic contrast check accessibility scores.
6	Other <i>(Maintainability)</i>	The source text formatting modules must be cleanly separated to allow regular algorithm refinement.	Code modules must adhere to strict documentation standards with a unit test coverage metric minimum of 80%.